

4.1.1 Energy changes in a system, and the ways energy is stored before and after such changes

4.1.1.1 Energy stores and systems

Content	Key opportunities for skills development
A system is an object or group of objects. There are changes in the way energy is stored when a system changes. Students should be able to describe all the changes involved in the way energy is stored when a system changes, for common situations. For example: • an object projected upwards • a moving object hitting an obstacle • an object accelerated by a constant force • a vehicle slowing down • bringing water to a boil in an electric kettle. Throughout this section on Energy students should be able to calculate the changes in energy involved when a system is changed by: • heating • work done by forces • work done when a current flows	The link between work done (energy transfer) and current flow in a circuit is covered in Energy transfers (page 29). WS 4.5
• use calculations to show on a common scale how the overall energy in a system is redistributed when the system is changed.	WS 1.2, 4.3, 4.5, 4.6 MS 1a, c, 3b, c

4.1.1.2 Changes in energy

Content	Key opportunities for skills development
Students should be able to calculate the amount of energy associated with a moving object, a stretched spring and an object raised above ground level.	WS 1.2, 4.3, 4.4, 4.6 MS 1a, c, 3b, c

Content	Key opportunities for skills development
The kinetic energy of a moving object can be calculated using the equation: $\text{kinetic energy} = 0.5 \times \text{mass} \times (\text{speed})^2$ $[E_k = \frac{1}{2} m v^2]$ kinetic energy, E_k, in joules, J mass, m, in kilograms, kg speed, v, in metres per second, m/s The amount of elastic potential energy stored in a stretched spring can be calculated using the equation:	MS 3b, c Students should be able to recall and apply this equation.
$\text{elastic potential energy} = 0.5 \times \text{spring constant} \times (\text{extension})^2$ $[E_e = \frac{1}{2} k e^2]$ (assuming the limit of proportionality has not been exceeded) elastic potential energy, E_e, in joules, J spring constant, k, in newtons per metre, N/m extension, e, in metres, m The amount of gravitational potential energy gained by an object raised above ground level can be calculated using the equation:	MS 3b, c Students should be able to apply this equation which is given on the <i>Physics equation sheet</i>.
$g.p.e. = \text{mass} \times \text{gravitational field strength} \times \text{height}$ $[E_p = m g h]$ gravitational potential energy, E_p, in joules, J mass, m, in kilograms, kg gravitational field strength, g, in newtons per kilogram, N/kg (In any calculation the value of the gravitational field strength (g) will be given). height, h, in metres, m	MS 3b, c Students should be able to recall and apply this equation. AT 1 Investigate the transfer of energy from a gravitational potential energy store to a kinetic energy store.

4.1.1.3 Energy changes in systems

Content	Key opportunities for skills development
The amount of energy stored in or released from a system as its temperature changes can be calculated using the equation:	

Content	Key opportunities for skills development
<i>change in thermal energy = mass × specific heat capacity × temperature change</i> $[\Delta E = m c \Delta \theta]$ change in thermal energy, ΔE, in joules, J mass, m, in kilograms, kg specific heat capacity, c, in joules per kilogram per degree Celsius, J/kg °C temperature change, $\Delta \theta$, in degrees Celsius, °C The specific heat capacity of a substance is the amount of energy required to raise the temperature of one kilogram of the substance by one degree Celsius.	MS 3b, c Students should be able to apply this equation which is given on the Physics equation sheet. This equation and specific heat capacity are also included in Temperature changes in a system and specific heat capacity (page 34).

Required practical activity 1: investigation to determine the specific heat capacity of one or more materials. The investigation will involve linking the decrease of one energy store (or work done) to the increase in temperature and subsequent increase in thermal energy stored.

AT skills covered by this practical activity: AT 1 and 5.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 89).

4.1.1.4 Power

Content	Key opportunities for skills development
Power is defined as the rate at which energy is transferred or the rate at which work is done. $power = \frac{energy\ transferred}{time}$ $\left[P = \frac{E}{t} \right]$ $power = \frac{work\ done}{time}$ $\left[P = \frac{W}{t} \right]$ power, P, in watts, W energy transferred, E, in joules, J time, t, in seconds, s work done, W, in joules, J An energy transfer of 1 joule per second is equal to a power of 1 watt. Students should be able to give examples that illustrate the definition of power eg comparing two electric motors that both lift the same weight through the same height but one does it faster than the other.	MS 3b, c Students should be able to recall and apply both equations.